Decimal Operations

Every graded set, every week — 5 weeks, 25 worksheets, with an answer key after each week.

WEEK 1
5 sets

WEEK 2
5 sets

WEEK 3
5 sets

WEEK 4
5 sets

WEEK 5
5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Adding Decimals

Line up the points, fill with zeros, then add as normal.

NAME __________

DATE __________

SCORE ____ / 11

✦ Warm-Up 6 ITEMS

0.3 + 0.4	1.2 + 2.5	0.25 + 0.25
3.5 + 1.5	0.6 + 0.9	2.75 + 1.25

◆ Core GRADED · SHOW YOUR WORKING

1. 3.5 + 0.47
2. 12.6 + 4.85
3. 0.125 + 0.875
4. 9.4 + 0.68
5. $14.75 + $8.60 — type the number

★ Challenge NOT GRADED

C1. 2.5 + 3.75 + 0.125

C2. What adds to 4.6 to make 10

Subtracting Decimals

Same alignment. Fill the gaps with zeros before you borrow.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

0.9 – 0.4	3.7 – 1.2	1 – 0.5
2.5 – 0.5	0.75 – 0.25	5.5 – 2.5

Core GRADED · SHOW YOUR WORKING

1. 4 – 1.35
2. 10 – 0.07
3. 8.2 – 3.45
4. $20 – $13.68 — type the number
5. 6.05 – 2.5

Challenge NOT GRADED

C1. 12.4 – 7.856

C2. A 2.5 m board with 0.85 m cut off — metres left

Multiplying Decimals

Multiply the digits, then count the decimal places.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up 6 ITEMS

0.5 × 4	0.2 × 6	1.5 × 2
0.25 × 4	3 × 0.1	0.5 × 10

Core GRADED · SHOW YOUR WORKING

1. 0.4 × 0.2 (Two decimal digits in, two out)
2. 1.2 × 0.5
3. 2.5 × 1.4
4. 0.06 × 40
5. 3.2 × 2.5

Challenge NOT GRADED

C1. 0.125 × 8

C2. 1.25 × 0.8

Dividing Decimals

Shift both numbers the same number of places, then divide as usual.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

4.8 ÷ 2	0.9 ÷ 3	6.4 ÷ 4
2.5 ÷ 5	0.36 ÷ 6	7.2 ÷ 8

Core GRADED · SHOW YOUR WORKING

1. 6 ÷ 0.5 (How many halves in six) ....
2. 4 ÷ 0.25 ....
3. 1.5 ÷ 0.5 ....
4. 9.6 ÷ 1.2 ....
5. $7.50 shared by 3 — type the number ....

Challenge NOT GRADED

C1. 0.144 ÷ 0.12

....

C2. A 4.5 m rope cut into 0.75 m pieces — how many

....

Unit Price Investigation

Enrichment. Price per unit is the only fair comparison.

NAME __________
DATE __________
SCORE ____ / 4

Warm-Up

2 ITEMS

$6 for 2 kg — price per kg

$10 for 5 L — price per litre

Core

GRADED · SHOW YOUR WORKING

1. $4.50 for 3 kg — price per kg
2. $7.20 for 0.8 kg — price per kg

Challenge

NOT GRADED

C1. Box A: $5 for 400 g. Box B: $9 for 800 g. Price per 100 g of B

C2. Same boxes — price per 100 g of A

C3. So which box is better value — type A or B

Week 1 Answers

Mission 04 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 · Adding Decimals

OUT OF 11

Warm-Up · 0.7 | 3.7 | 0.5 | 5 | 1.5 | 4

Core · 3.97 | 17.45 | 1 | 10.08 | 23.35

Challenge · 6.375 | 5.4

1.2 · Subtracting Decimals

OUT OF 11

Warm-Up · 0.5 | 2.5 | 0.5 | 2 | 0.5 | 3

Core · 2.65 | 9.93 | 4.75 | 6.32 | 3.55

Challenge · 4.544 | 1.65

1.3 · Multiplying Decimals

OUT OF 11

Warm-Up · 2 | 1.2 | 3 | 1 | 0.3 | 5

Core · 0.08 | 0.6 | 3.5 | 2.4 | 8

Challenge · 1 | 1

1.4 · Dividing Decimals

OUT OF 11

Warm-Up · 2.4 | 0.3 | 1.6 | 0.5 | 0.06 | 0.9

Core · 12 | 16 | 3 | 8 | 2.5

Challenge · 1.2 | 6

Fri · Unit Price Investigation

OUT OF 4

Warm-Up · 3 | 2

Core · 1.5 | 9

Challenge · 1.125 | 1.25 | B

NAME __________
DATE __________
SCORE ____ / 11

Ignore the Point

Multiply the digits first, then count the places.

Warm-Up
6 ITEMS

5 × 4	0.5 × 4	12 × 5
1.2 × 5	25 × 4	0.25 × 4

Core
GRADED · SHOW YOUR WORKING

1. 0.4 × 0.2
2. 1.2 × 0.5
3. 2.5 × 1.4
4. 0.06 × 40
5. 3.2 × 2.5

Challenge
NOT GRADED

C1. 0.125 × 8

C2. 1.25 × 0.8

Count the Places

Two decimal digits in, two decimal digits out.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Decimal digits in 0.4	In 0.25	In 1.234
0.2 × 0.3	0.5 × 0.2	0.1 × 0.1

Core

GRADED · SHOW YOUR WORKING

1. 0.25 × 0.4
2. 1.5 × 0.02
3. 0.12 × 0.5
4. 2.4 × 0.25
5. 0.75 × 0.8

Challenge

NOT GRADED

C1. 0.125 × 0.4

C2. 1.25 × 1.6

Place It by Estimating

0.4 × 60 is about 24, so the point can only go one place.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Estimate 0.5 × 40	0.5 × 40	Estimate 2 × 30
1.9 × 30	Estimate 0.25 × 80	0.25 × 80

Core

GRADED · SHOW YOUR WORKING

1. Estimate 0.4 × 60
2. 0.4 × 62
3. Estimate 4.9 × 21
4. 4.9 × 20
5. Estimate 0.9 × 150

Challenge

NOT GRADED

C1. 0.75 × 240

C2. Estimate 1.02 × 500

Smaller Than You Started

Why × 0.5 halves instead of doubling.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up

6 ITEMS

10 × 0.5	10 × 2	10 × 1
8 × 0.5	20 × 0.25	6 × 0.5

Core

GRADED · SHOW YOUR WORKING

1. 40 × 0.75 — bigger or smaller than 40
2. 40 × 0.75
3. 40 × 1.25 — bigger or smaller
4. 40 × 1.25
5. 100 × 0.1

Challenge

NOT GRADED

C1. Multiplying by a number under 1 makes it — type bigger or smaller

C2. 60 × 0.05

Point Placement

Given the digits, race to place the decimal point.

NAME __________

DATE __________

SCORE ____ / 4

Warm-Up

2 ITEMS

Digits 24, one decimal place — the number

Digits 24, two places

Core

GRADED · SHOW YOUR WORKING

1. 0.6 × 0.4 — digits 24, so the answer

2. 6 × 0.4 — the answer

Challenge

NOT GRADED

C1. 1.5 × 0.16 — digits 240, so the answer

C2. 0.15 × 1.6

C3. 15 × 0.016

Week 2 Answers

Mission 04 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 · Ignore the Point

OUT OF 11

Warm-Up · 20 | 2 | 60 | 6 | 100 | 1

Core · 0.08 | 0.6 | 3.5 | 2.4 | 8

Challenge · 1 | 1

2.2 · Count the Places

OUT OF 11

Warm-Up · 1 | 2 | 3 | 0.06 | 0.1 | 0.01

Core · 0.1 | 0.03 | 0.06 | 0.6 | 0.6

Challenge · 0.05 | 2

2.3 · Place It by Estimating

OUT OF 11

Warm-Up · 20 | 20 | 60 | 57 | 20 | 20

Core · 24 | 24.8 | 100 | 98 | 135

Challenge · 180 | 500

2.4 · Smaller Than You Started

OUT OF 11

Warm-Up · 5 | 20 | 10 | 4 | 5 | 3

Core · smaller | 30 | bigger | 50 | 10

Challenge · smaller | 3

Fri · Point Placement

OUT OF 4

Warm-Up · 2.4 | 0.24

Core · 0.24 | 2.4

Challenge · 0.24 | 0.24 | 0.24

Decimal ÷ Whole

The point comes straight up. Nothing else moves.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up 6 ITEMS

4.8 ÷ 2	0.9 ÷ 3	6.4 ÷ 4
2.5 ÷ 5	0.36 ÷ 6	7.2 ÷ 8

Core GRADED · SHOW YOUR WORKING

1. 9.6 ÷ 4 ....
2. 1.44 ÷ 12 ....
3. 5.25 ÷ 5 ....
4. 8.4 ÷ 7 ....
5. 0.72 ÷ 9 ....

Challenge NOT GRADED

C1. 12.5 ÷ 4

C2. 7.35 ÷ 15

Whole ÷ Decimal

Shift both the same number of places, then divide as usual.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up 6 ITEMS

$6 \div 0.5$	$4 \div 0.5$	$10 \div 0.5$
$4 \div 0.25$	$3 \div 0.5$	$2 \div 0.25$

Core GRADED · SHOW YOUR WORKING

1. $9 \div 0.3$
2. $12 \div 0.4$
3. $5 \div 0.125$
4. $7 \div 0.7$
5. $20 \div 0.8$

Challenge NOT GRADED

C1. $15 \div 0.06$

C2. $1 \div 0.004$

Decimal ÷ Decimal

Same shift, applied to both numbers.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

0.6 ÷ 0.2	0.8 ÷ 0.4	1.5 ÷ 0.5
0.9 ÷ 0.3	2.4 ÷ 0.6	1.2 ÷ 0.4

Core GRADED · SHOW YOUR WORKING

1. 9.6 ÷ 1.2
2. 0.144 ÷ 0.12
3. 4.5 ÷ 1.5
4. 6.25 ÷ 2.5
5. 0.81 ÷ 0.09

Challenge NOT GRADED

C1. 7.5 ÷ 0.25

C2. 0.0144 ÷ 0.012

Money Problems

Unit prices and change, to the cent.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

$6 for 2 — each	$10 for 4 — each	$9 for 3 — each
$20 – $12.50	$5 ÷ 4	3 at $2.50

Core

GRADED · SHOW YOUR WORKING

1. $7.50 shared by 3
2. $4.50 for 3 kg — per kg
3. $7.20 for 0.8 kg — per kg
4. $20 – 3 at $4.99
5. $15 ÷ 12

Challenge

NOT GRADED

C1. $5 for 400 g — per 100 g

C2. $9 for 800 g — per 100 g

Unit Price Investigation Begins

Collect real prices and sizes. No conclusions yet.

NAME __________
DATE __________
SCORE ____ / 4

Warm-Up
2 ITEMS

$8 for 4 — each

$12 for 6 — each

Core
GRADED · SHOW YOUR WORKING

1. $3.60 for 4 — each
2. $11.25 for 5 — each

Challenge
NOT GRADED

C1. 1 kg at $8, or 750 g at $5.40 — per kg for the small one

C2. Which is better value — type 1kg or 750g

C3. $2.40 for 6 — each

TEACHER PAGE · ANSWER KEY

Week 3 Answers

Mission 04 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 · Decimal ÷ Whole

OUT OF 11

Warm-Up · 2.4 | 0.3 | 1.6 | 0.5 | 0.06 | 0.9

Core · 2.4 | 0.12 | 1.05 | 1.2 | 0.08

Challenge · 3.125 | 0.49

3.2 · Whole ÷ Decimal

OUT OF 11

Warm-Up · 12 | 8 | 20 | 16 | 6 | 8

Core · 30 | 30 | 40 | 10 | 25

Challenge · 250 | 250

3.3 · Decimal ÷ Decimal

OUT OF 11

Warm-Up · 3 | 2 | 3 | 3 | 4 | 3

Core · 8 | 1.2 | 3 | 2.5 | 9

Challenge · 30 | 1.2

3.4 · Money Problems

OUT OF 11

Warm-Up · 3 | 2.5 | 3 | 7.5 | 1.25 | 7.5

Core · 2.5 | 1.5 | 9 | 5.03 | 1.25

Challenge · 1.25 | 1.125

Fri · Unit Price Investigation Begins

OUT OF 4

Warm-Up · 2 | 2

Core · 0.9 | 2.25

Challenge · 7.2 | 750g | 0.4

Choose the Operation

The words tell you. Read them twice before reaching for a method.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

3 items at \$2.50 — total	\$10 shared by 4	\$5.50 + \$2.50
\$10 – \$3.25	4 at \$1.25	\$9 ÷ 3

Core GRADED · SHOW YOUR WORKING

1. 2.5 kg at \$4 per kg
2. \$12 buys how many at \$2.40
3. 1.5 m plus 0.85 m — metres
4. 6 at \$3.99 — total
5. \$50 – \$27.45

Challenge NOT GRADED

C1. 0.75 kg at \$6.40 per kg

C2. \$18 buys how many at \$0.75

Two-Step Problems

Multiply then subtract, in that order.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up

6 ITEMS

3 at $2, change from $10	4 at $1.50, change from $10	2 at $3.25 — total
Change from $10	5 at $2 — total	Change from $20

Core

GRADED · SHOW YOUR WORKING

1. 3 at $4.99, change from $20
2. 6 at $1.75 — total
3. Change from $20
4. 2.5 kg at $3.20, change from $10
5. 4 at $6.25 — total

Challenge

NOT GRADED

C1. $40 less 3 dinners at $11.25

C2. 1.5 kg at $4.80 plus $2.50 delivery

Measurement Contexts

Metres, litres and kilograms, all carrying decimals.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

1.5 m + 0.5 m	2.5 L − 1 L	0.5 kg × 4
3 m ÷ 2	1.2 m + 0.8 m	4.5 kg − 1.5 kg

Core

GRADED · SHOW YOUR WORKING

1. 1.25 m × 4
2. 7.5 L shared into 0.5 L bottles — how many
3. 2.4 kg ÷ 3
4. 0.75 m × 6
5. 12.5 L − 4.75 L

Challenge

NOT GRADED

C1. A 4.5 m rope into 0.75 m pieces — how many

C2. 3.6 kg shared by 8 — kg each

Estimate Everything First

Write the estimate before the working, every time.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Estimate 4.9 + 3.1	Estimate 9.8 – 4.9	Estimate 2.1 × 4
Estimate 11.9 ÷ 4	Estimate 0.9 + 1.1	Estimate 5.2 × 2

Core

GRADED · SHOW YOUR WORKING

1. Estimate 12.4 – 7.9
2. True value of 12.4 – 7.856
3. Estimate 0.49 × 100
4. True value of 0.49 × 100
5. Estimate 19.8 ÷ 5

Challenge

NOT GRADED

C1. Estimate 6 items at $4.95

C2. True cost of 6 at $4.95

Mid-Unit Quiz

Eight items across Weeks 1–4. 85% to keep flying.

NAME __________

DATE __________

SCORE ____ / 7

Warm-Up 2 ITEMS

0.3 + 0.4

0.5 × 4

Core GRADED · SHOW YOUR WORKING

1. 3.5 + 0.47
2. 8.2 – 3.45
3. 0.4 × 0.2
4. 6 ÷ 0.5
5. $7.20 for 0.8 kg — per kg

Challenge NOT GRADED

C1. 12.4 – 7.856

Week 4 Answers

Mission 04 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 · Choose the Operation

OUT OF 11

Warm-Up · 7.5 | 2.5 | 8 | 6.75 | 5 | 3

Core · 10 | 5 | 2.35 | 23.94 | 22.55

Challenge · 4.8 | 24

4.2 · Two-Step Problems

OUT OF 11

Warm-Up · 4 | 4 | 6.5 | 3.5 | 10 | 10

Core · 5.03 | 10.5 | 9.5 | 2 | 25

Challenge · 6.25 | 9.7

4.3 · Measurement Contexts

OUT OF 11

Warm-Up · 2 | 1.5 | 2 | 1.5 | 2 | 3

Core · 5 | 15 | 0.8 | 4.5 | 7.75

Challenge · 6 | 0.45

4.4 · Estimate Everything First

OUT OF 11

Warm-Up · 8 | 5 | 8 | 3 | 2 | 10

Core · 4 | 4.544 | 50 | 49 | 4

Challenge · 30 | 29.7

Fri · Mid-Unit Quiz

OUT OF 7

Warm-Up · 0.7 | 2

Core · 3.97 | 4.75 | 0.08 | 12 | 9

Challenge · 4.544

Compute Unit Prices

Divide price by size for every item on the list.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up
6 ITEMS

$6 for 2 kg — per kg	$10 for 5 L — per L	$8 for 4 — each
$12 for 3 — each	$15 for 5 — each	$9 for 3 — each

Core
GRADED · SHOW YOUR WORKING

1. $4.50 for 3 kg — per kg
2. $7.20 for 0.8 kg — per kg
3. $11.25 for 5 — each
4. $2.40 for 6 — each
5. $13.50 for 1.5 kg — per kg

Challenge
NOT GRADED

C1. $5 for 400 g — per 100 g

C2. $9 for 800 g — per 100 g

NAME __________
DATE __________
SCORE ____ / 11

Find the Trap

Name one product where the bigger box costs more per unit.

Warm-Up

6 ITEMS

$4 for 2 — each	$9 for 5 — each	Cheaper each — type 1.8 or 2
$6 for 3 — each	$10 for 4 — each	Cheaper each — type 2 or 2.5

Core

GRADED · SHOW YOUR WORKING

1. 400 g at $5 — per 100 g
2. 800 g at $9 — per 100 g
3. Better value — type 400 or 800
4. 500 g at $4 — per 100 g
5. 1 kg at $9 — per 100 g

Challenge

NOT GRADED

C1. So which is better value — type 500g or 1kg

C2. The saving per 100 g by picking it

Write the Recommendation

One paragraph a shopper could act on.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

$1.25 per 100 g — the cost of 400 g	$0.90 per 100 g — the cost of 1 kg	$2 each — the cost of 5
$1.80 each — the cost of 5	The saving	$3 per kg — the cost of 2 kg

Core GRADED · SHOW YOUR WORKING

1. Saving $0.125 per 100 g on 800 g
2. Buying 800 g weekly — the yearly saving over 52 weeks
3. $1.25 vs $1.125 per 100 g — the percent saved, to the nearest whole
4. $9 vs $10 — the percent saved
5. Saving $1 a week for a year

Challenge NOT GRADED

C1. A $0.10 per 100 g saving on 2 kg weekly, over 52 weeks

C2. Which matters more for a weekly buy — type per-unit or total

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up 6 ITEMS

0.25 + 0.25	1 – 0.5	0.5 × 4
4.8 ÷ 2	0.6 + 0.9	$10 for 5 — each

Core GRADED · SHOW YOUR WORKING

1. 12.6 + 4.85
2. 10 – 0.07
3. 0.4 × 0.2
4. 9.6 ÷ 1.2
5. 2.5 × 1.4

Challenge NOT GRADED

C1. 12.4 – 7.856

C2. 1.25 × 0.8

Mission 04 Test

Twelve items plus the Big Question, answered out loud.

NAME __________
DATE __________
SCORE ____ / 10

Warm-Up

2 ITEMS

1.2 + 2.5

0.9 – 0.4

Core

GRADED · SHOW YOUR WORKING

1. 3.5 + 0.47
2. 4 – 1.35
3. 1.2 × 0.5
4. 3.2 × 2.5
5. 6 ÷ 0.5
6. 9.6 ÷ 1.2
7. $4.50 for 3 kg — per kg
8. 3 at $4.99 — change from $20

Challenge

NOT GRADED

C1. 12.4 – 7.856

C2. A 4.5 m rope into 0.75 m pieces — how many

Week 5 Answers

Mission 04 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

5.1 · Compute Unit Prices

OUT OF 11

Warm-Up · 3 | 2 | 2 | 4 | 3 | 3

Core · 1.5 | 9 | 2.25 | 0.4 | 9

Challenge · 1.25 | 1.125

5.2 · Find the Trap

OUT OF 11

Warm-Up · 2 | 1.8 | 1.8 | 2 | 2.5 | 2

Core · 1.25 | 1.125 | 800 | 0.8 | 0.9

Challenge · 500g | 0.1

5.3 · Write the Recommendation

OUT OF 11

Warm-Up · 5 | 9 | 10 | 9 | 1 | 6

Core · 1 | 52 | 10 | 10 | 52

Challenge · 104 | per-unit

Thu · Error Journal Sweep

OUT OF 11

Warm-Up · 0.5 | 0.5 | 2 | 2.4 | 1.5 | 2

Core · 17.45 | 9.93 | 0.08 | 8 | 3.5

Challenge · 4.544 | 1

Fri · Mission 04 Test

OUT OF 10

Warm-Up · 3.7 | 0.5

Core · 3.97 | 2.65 | 0.6 | 8 | 12 | 8 | 1.5 | 5.03

Challenge · 4.544 | 6